Practical No. 5

Aim: Write a java program demonstrating inheritance implementation.

Code:

Program 1:

```
// Base Class / Parent Class
class Vehicle {
    protected String brand = "Ford";

    public void honk() {
        System.out.println("Beep, beep!");
    }
}

// Derived Class / Child Class (Single Inheritance)
class Car extends Vehicle {
    private String modelName = "Mustang";

    public void displayInfo() {
        // Inherits 'brand' field and 'honk()' method from Vehicle
        System.out.println("Brand: " + brand);
        System.out.println("Model: " + modelName);
    }
}

public class SingleInheritanceDemo {
    public static void main(String[] args) {
        Car myCar = new Car();

        // Call method inherited from Vehicle
        myCar.honk();

        // Call method defined in Car
        myCar.displayInfo();
    }
}}
```

Program 2:

```
// Define the Interface
interface Playable {
    // Interface methods are implicitly 'public' and 'abstract'
    void play();
    void pause();
    void stop();
}

// Class implementing the Interface
class AudioPlayer implements Playable {
    private String songTitle;

    public AudioPlayer(String songTitle) {
        this.songTitle = songTitle;
```

```
    }

    // Must provide public implementations for all interface methods
    @Override
    public void play() {
        System.out.println("Playing audio: " + songTitle);
    }

    @Override
    public void pause() {
        System.out.println("Paused audio: " + songTitle);
    }

    @Override
    public void stop() {
        System.out.println("Stopped playing: " + songTitle);
    }
}

public class BasicInterfaceDemo {
    public static void main(String[] args) {
        // Polymorphic reference: Interface type holding an instance of the implementing class
        Playable player = new AudioPlayer("Bohemian Rhapsody");

        player.play();
        player.pause();
        player.stop();
    }
}
```

Output:

```
Beep, beep!
Brand: Ford
Model: Mustang
```

```
Playing audio: Bohemian Rhapsody
Paused audio: Bohemian Rhapsody
Stopped playing: Bohemian Rhapsody
```